
* Core Novelty of Maa's over agent based & statistical based models?

- LMM is generative foundational model. with finest grained order level by autoregressive generation of "trading orders" & "order batches".
- They are different from
 - a) Statistical (Geometric Brownian Motion (or) GARCH for volatility modeling)
 - b) Agent based (ABM with predefined heterogeneous behaviour & rules).

- Statistical:-

They rely on parametric assumption about market distribution.

Eg:- log-normal returns.

• price time series & volumes.

Often fitted by regression or MCS (Monte Carlo Simulation).

Simple ✓
Compute efficient ✓

Realism X
(to capture micro level dynamics),

- individual order placements
- cancellations
- ripple effects
- "Second order effects"

large buy order will trigger cascading sell orders.

- Agent based modeling (ABM).

Improvement over the statistical method. by modeling interactions based on "rule based agents"

- requires specifying behaviour
- hyperparameters
- interaction rules.

Leading to brittleness in the modeling.

- Maa's (LMM)

- Transformer based generative model.
- Trained over vast order level historical data (NASDAQ ITCH)
millisecond time steps.

→ Unlike statistical models which uses top down probabilistic fitting (or) ABM bottom up rule setting.

LMM is trained to predict next token based on the data.
 → High Fidelity

this replicates market simulation. (market micro structure).

→ also includes LOB (upto 10-level bid/ask depth).

☆☆ (Able to reproduce)
 → 11 stylized facts
 ↳ out of which 9 are present.

Fact #	Fact Name	Historical	Simulated
1	Absence of autocorrelations	x	x
2	Heavy tails	x	x
→ 3	Gain/loss asymmetry		
4	Aggregational Gaussianity	x	x
5	Intermittency	x	x
6	Volatility clustering	x	x
7	Conditional heavy tails	x	x
8	Slow decay of autocorrelation in absolute returns	x	x
→ 9	Leverage effect		
10	Volume/volatility correlation	x	x
11	Asymmetry in timescales	x	x

Table 6: Presence of Stylized Facts in Historical and Simulated Order Sequences. All facts are present in both historical and simulated data, except for *Gain/loss asymmetry* and *Leverage effect*.

→ Transformer size is around - 3B parameter able to capture complex non linear dependencies without explicit rules.

→ The paper also implemented "Simulated Clearing house" where the user can inject order in real time and see the dynamics.

Note:- Read about Square Root law:



Probability (Statistics) → Rule based (ABM) → Descriptive (LMM) (Data driven)

- Realism
- interactivity.
- order blending.
- agent training.

